

Volume of Cuboids



What is a Cuboid?

A cuboid is a 3D shape with 6 rectangular faces. You can find cuboids everywhere - books, boxes, bricks, and even rooms!





Parts of a Cuboid

A cuboid has three important measurements:

- Length (l) - how long it is
- Width (w) or Breadth (b) - how wide it is
- Height (h) - how tall it is

What is Volume?

Volume is the amount of space inside a 3D object. We measure volume in cubic centimeters (cm^3) or cubic meters (m^3). Think of it this way: volume tells us how much water can fit inside a tank or how many small cubes can fill a box!

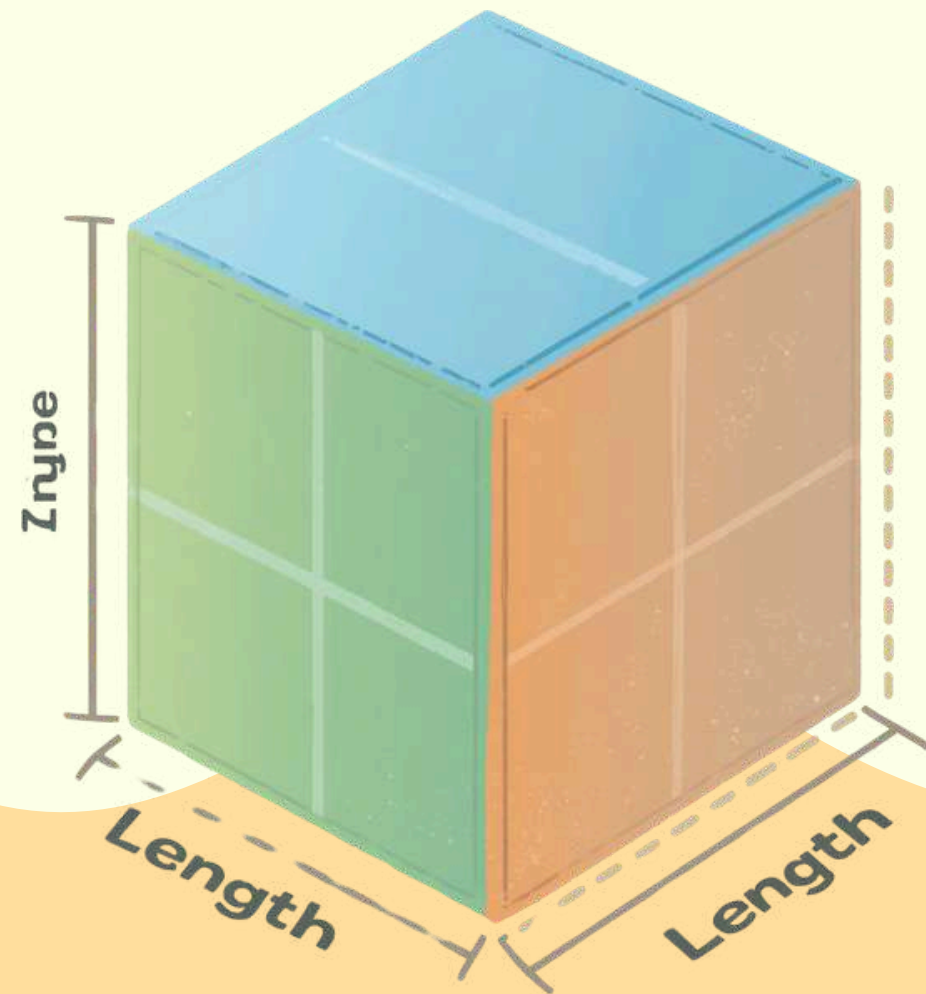


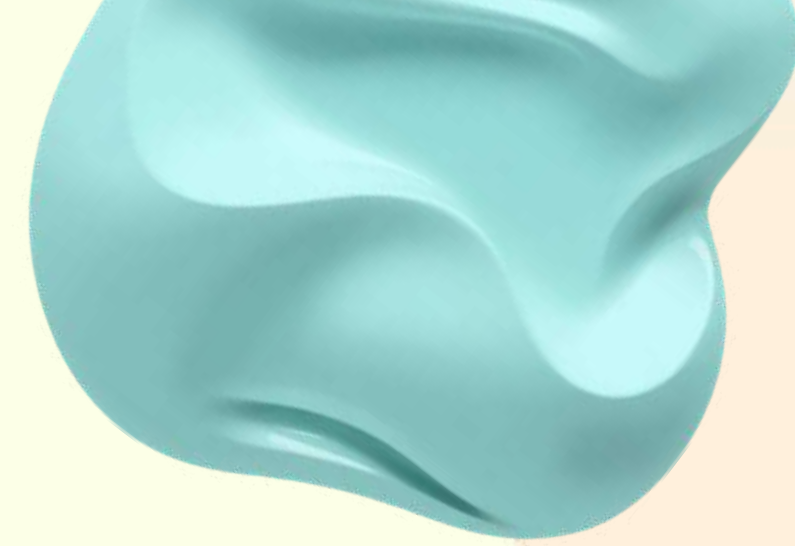
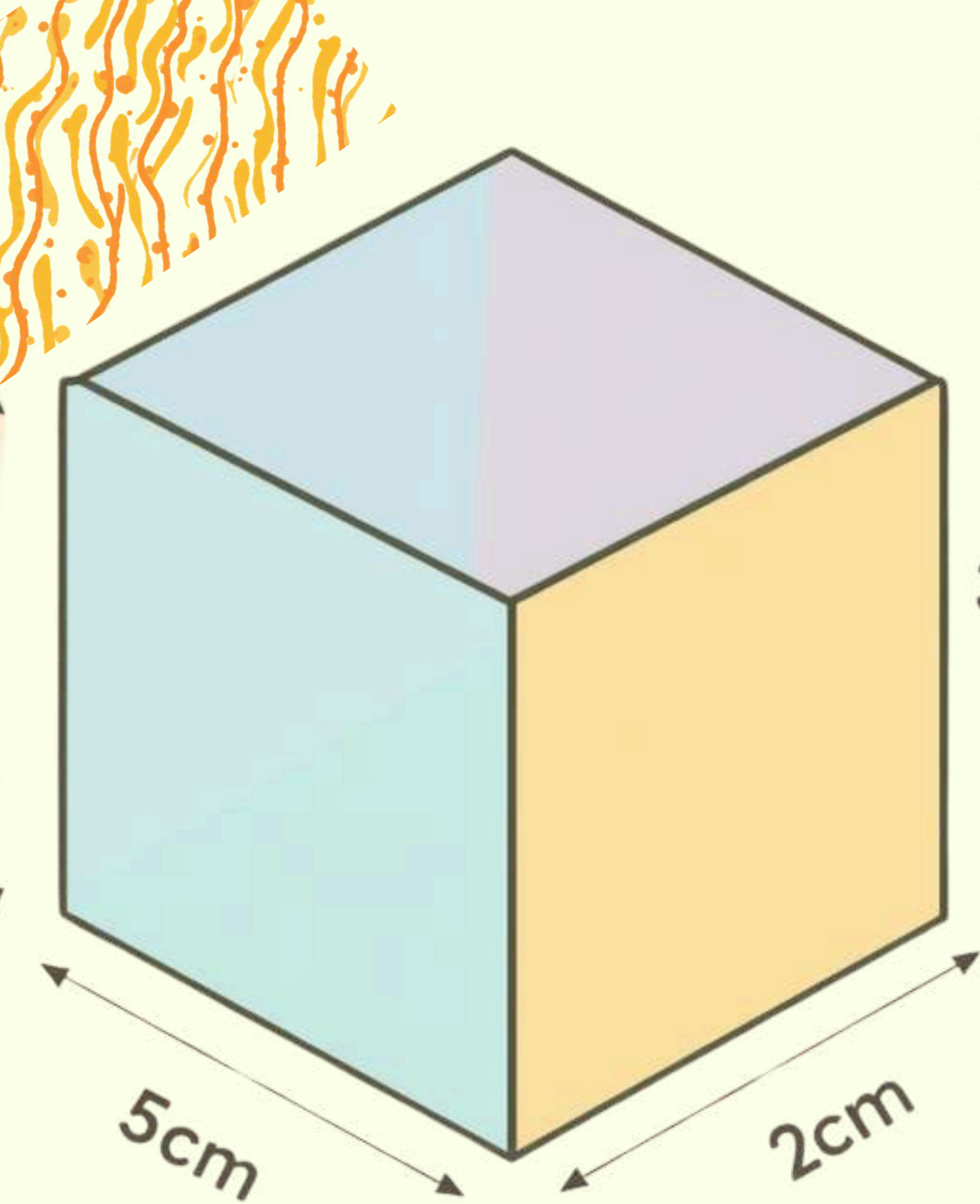
Volume Formula

Volume = Length × Width × Height

$$V = l \times w \times h$$

Multiply all three dimensions to find the volume of any cuboid!





Example 1 - Simple Calculation

Let's calculate the volume of a small box!

Dimensions: Length = 5 cm, Width = 3
cm, Height = 2 cm

$$V = 5 \times 3 \times 2$$

$$V = 30 \text{ cm}^3$$



Real-Time Example: Pencil Box

Dimensions: 20 cm × 8 cm × 5 cm

$$V = 20 \times 8 \times 5 = 800 \text{ cm}^3$$

This means the pencil box can hold 800
small cubes of 1 cm^3 each!





Example 2 - Classroom Scenario

Let's calculate the volume of a school desk!

Dimensions:

Length = 60 cm, Width = 45 cm, Height = 75
cm

$$V = 60 \times 45 \times 75$$

$$V = 202,500 \text{ cm}^3$$



Real-Time Example: Lunch Box

Indian Tiffin Box Dimensions:

Length = 15 cm, Width = 10 cm, Height = 6 cm

Volume = $15 \times 10 \times 6 = 900 \text{ cm}^3$

This tells us how much food the box can hold!

Example 3 - Water Tank

Tank Dimensions: 2 m × 1.5 m × 1 m

$$V = 2 \times 1.5 \times 1 = 3 \text{ m}^3$$

$$3 \text{ m}^3 = 3,000 \text{ liters}$$

That's enough water for a family's daily use!





78⁹²

Real-Time Example 3 - Book

Textbook Dimensions:

Length = 25 cm, Width = 18 cm, Height = 2 cm

Volume = $25 \times 18 \times 2 = 900 \text{ cm}^3$

That's how much space your textbook takes up!

Practice Problem 1

Find the volume of a gift box with these dimensions:

Length = 12 cm

Width = 10 cm

Height = 8 cm

Answer: $V = 12 \times 10 \times 8 = 960 \text{ cm}^3$



Practice Problem 2

Calculate the volume of a brick with these dimensions:

Length = 20 cm

Width = 10 cm

Height = 7 cm

Answer: $V = 20 \times 10 \times 7 = 1,400 \text{ cm}^3$



Real-Life Applications

Volume helps us in daily life! We use it for packing boxes when moving, calculating room size for furniture, finding aquarium water capacity, and organizing storage containers.



Units of Volume

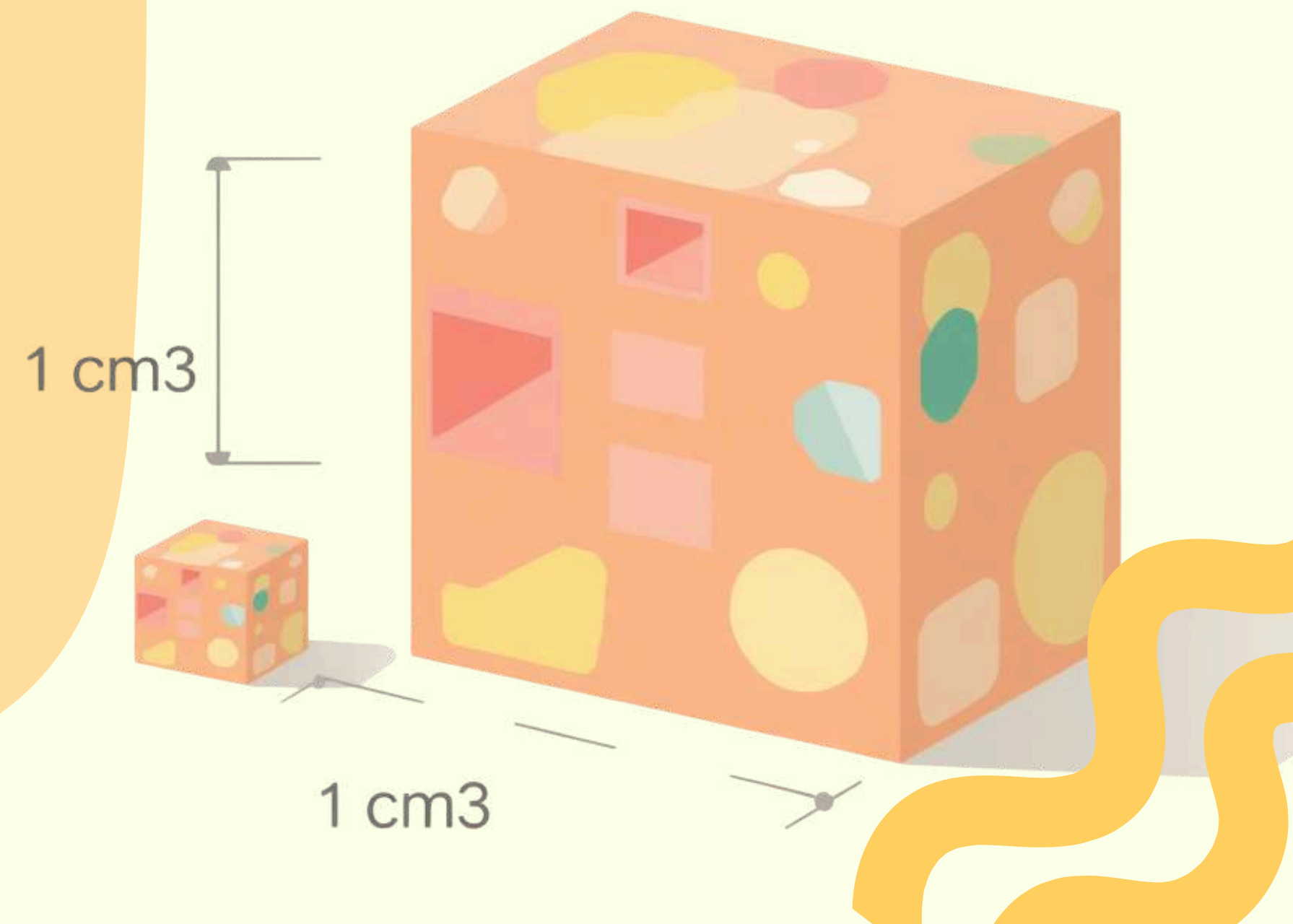
We measure volume in cubic units:

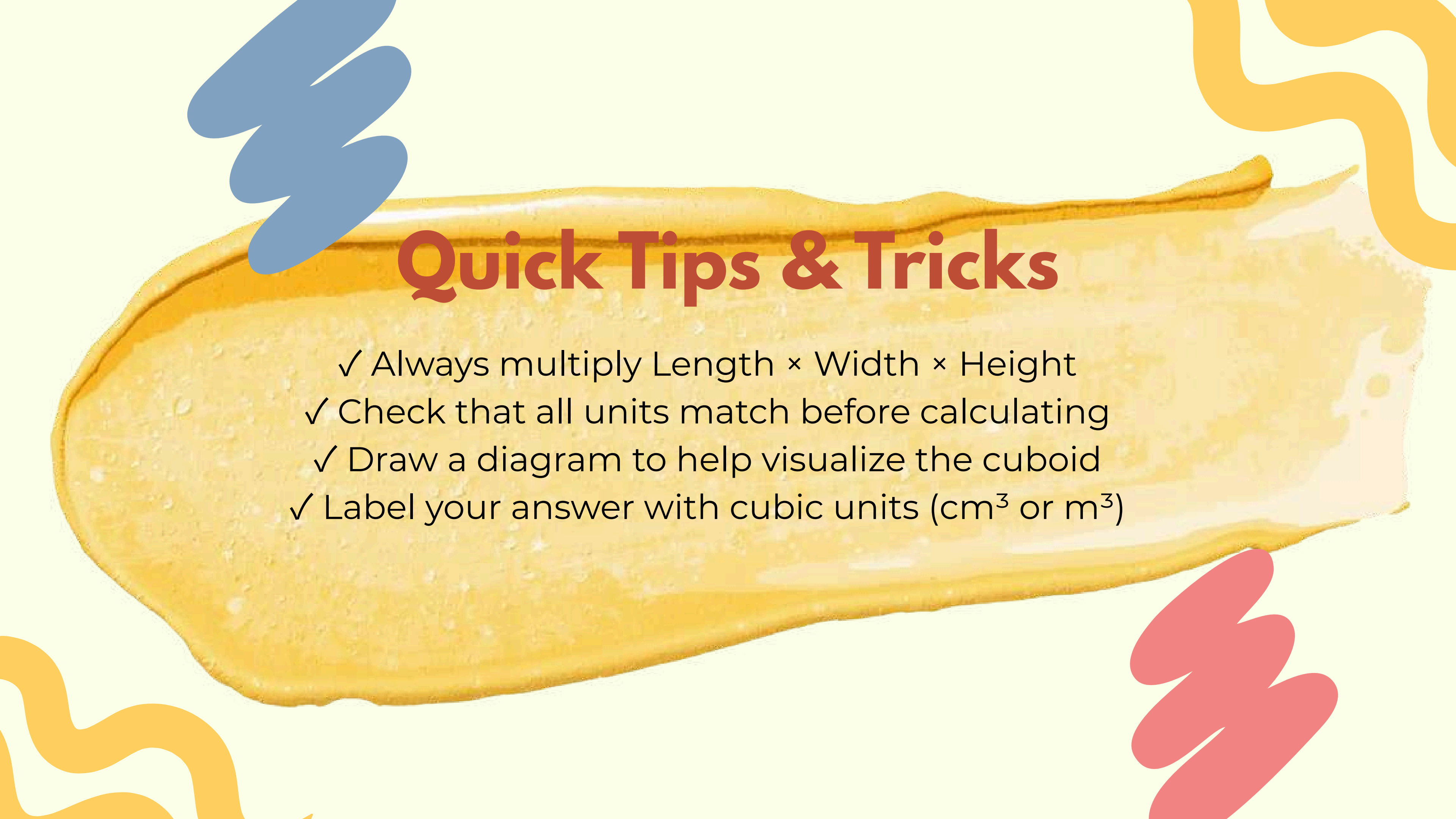
- Cubic centimeters (cm^3) - for small objects like boxes and books
- Cubic meters (m^3) - for large spaces like rooms and tanks

Conversion: $1 \text{ m}^3 = 1,000,000 \text{ cm}^3$

Volume Chart

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Quick Tips & Tricks

- ✓ Always multiply Length $\times$ Width $\times$ Height
- ✓ Check that all units match before calculating
- ✓ Draw a diagram to help visualize the cuboid
- ✓ Label your answer with cubic units (cm^3 or m^3)

Fun Challenge

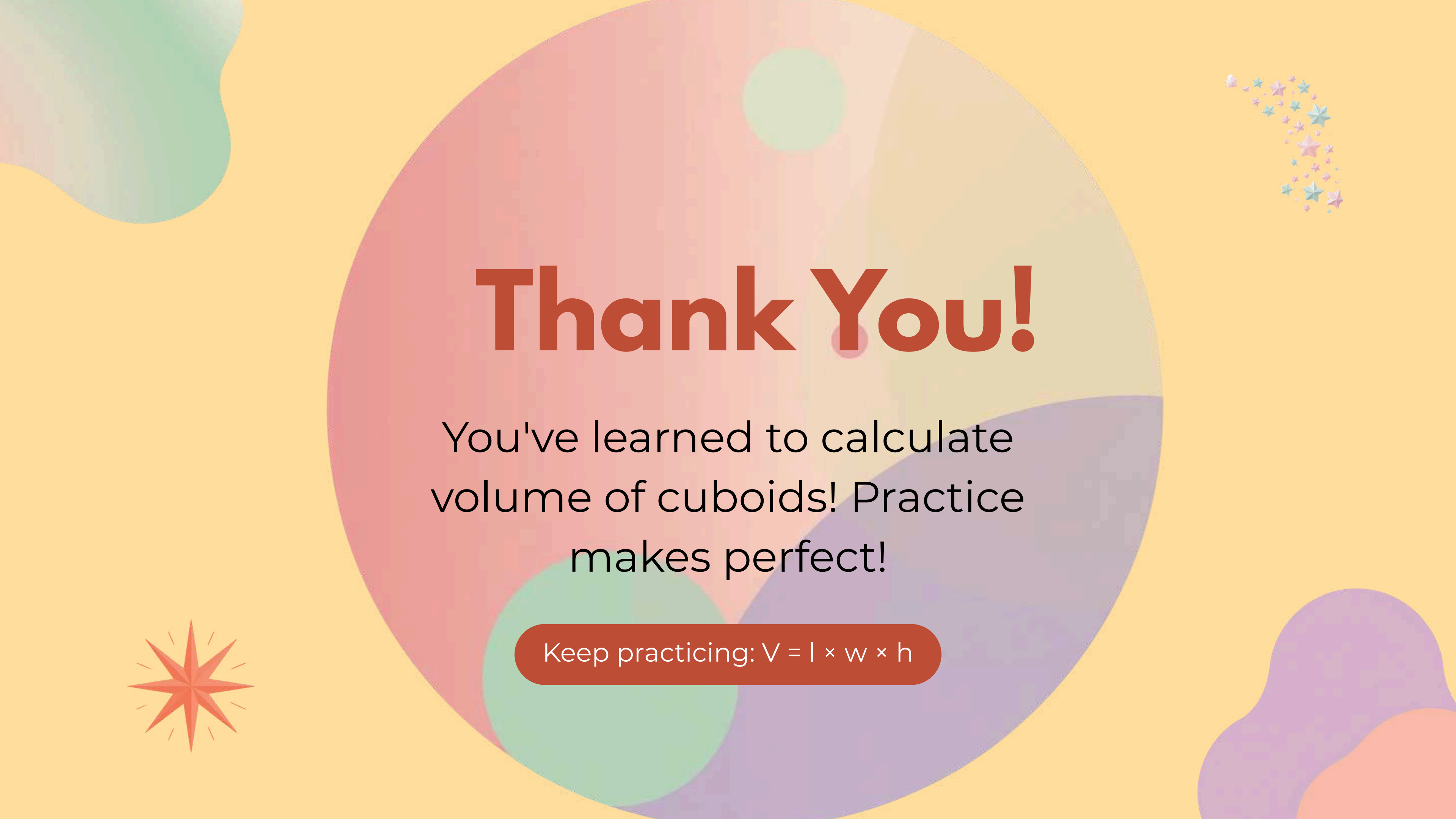
Mystery Box Problem:

Volume = 240 cm^3 , Length = 8 cm, Width = 6 cm

Find the Height (h)?

Solution: $h = 240 \div (8 \times 6) = 240 \div 48 = 5 \text{ cm}$





Thank You!

You've learned to calculate
volume of cuboids! Practice
makes perfect!

Keep practicing: $V = l \times w \times h$